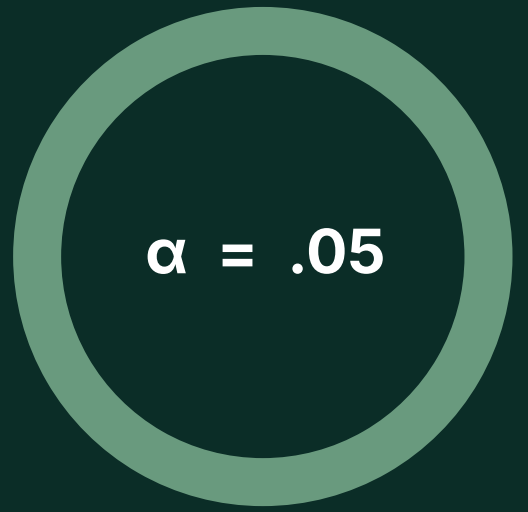
SHOPNOW • ANALYTICS PRESENTATION

What does the data really tell us?

Statistical testing, customer behaviour and business decisions



10
statistical questions

3
business challenges

1
important lesson

Signal
vs.
noise

TANVI MISHRA / PROJECT PRESENTATION

Spending gaps are visible — not decisive

One-sided or two-sided Welch t-tests, as appropriate • Significance threshold: p < 0.05

COMPARISON	GROUP A	GROUP B	P-VALUE
UK vs Germany	$234.84	$239.03	0.805
Discounted vs not	$235.61	$234.56	0.688
Male vs female	$234.81	$235.94	0.698
Email vs social	$236.82	$232.19	0.107

Interpretation: none of these comparisons is statistically significant at 5%.

Data source: ShopNow analysis, Q1–Q4. Values represent average purchase value per transaction.

Conversion is similar across device types

Chi-square independence test • $\chi^2 = 1.15$ • $p = 0.563$

CONVERSION RATE BY DEVICE



KEY RESULT

No detected device association

ADDITIONAL CHECK

Guest: 13.13%
Logged in: 13.07%

Login test p = 0.841

Implication: improve the shopping journey across devices and login states.

Browsing more does not imply spending more

Pearson correlation tested linear relationships in the ShopNow data

TIME ON SITE ↔ PURCHASE VALUE

+0.0094

Pearson r

p = 0.350

Essentially zero linear association

PAGES VIEWED ↔ PURCHASE VALUE

−0.0183

Pearson r

p = 0.069

Very weak negative association

TIME ON SITE ↔ PAGES VIEWED

+0.0045

Pearson r

p = 0.229

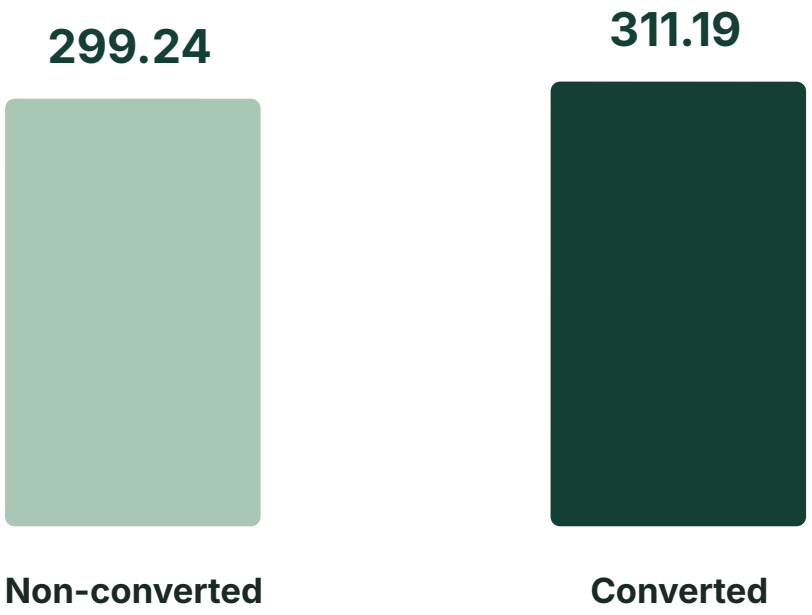
Essentially zero linear association

Next step: test buyer intent, product mix, checkout friction and repeat behaviour.

Significant does not mean important

Challenge 3 • Converted and non-converted session time

AVERAGE TIME ON SITE / DATASET UNITS



STATISTICAL TEST

p < 0.001

EFFECT SIZE

Cohen's d = 0.10

11.96 units

Difference in average time: statistically detectable, but a very small standardized effect.

Takeaway: judge effect size, cost and business outcomes — not just p-values.

From statistical tests to better decisions

What should ShopNow investigate next?

01

Focus on business outcomes

Track conversion alongside contribution margin, acquisition cost and customer lifetime value.

02

Validate before segmenting

Standardize categories (e.g., Email/email) and confirm differences before targeting.

03

Test intent, not just clicks

Explore product interest, repeat purchasing and checkout friction through controlled experiments.

DISCUSSION

What matters more than a small p-value?

Which follow-up analysis would be most useful to the business?

PROJECT TOOLS: Python • Pandas • SciPy • Matplotlib • Jupyter Notebook.